

CEREBRAL PALSY

Physiotherapy Assessment & Management

Introduction :

- It is a group of disorder of posture and movement that is secondary to the damage of developing fetal or infant brain
- This is a non-progressive disorder
- It involves a broad spectrum of motor disability
- Apart from the motor disability, there may be problem associated with:
 - Cognition
 - Seizure
 - Vision
 - Speech
 - Swallowing
 - Orthopedic deformities

Etiology and Risk Factors :

- **Pre-Natal**
 - TORCH infection
 - Hypoglycemia
 - RH incompatibility
 - Placental abnormalities
 - Drug side effect
- **Peri-Natal**
 - Pre-Term delivery
 - ICA/MCA occlusion during birth
 - Trauma
 - Asphyxia
 - Neonatal Meningitis



Etiology and Risk Factors :

- **Post-Natal**
 - Delayed cry
 - Jaundice
 - Trauma
 - Infection



Classification

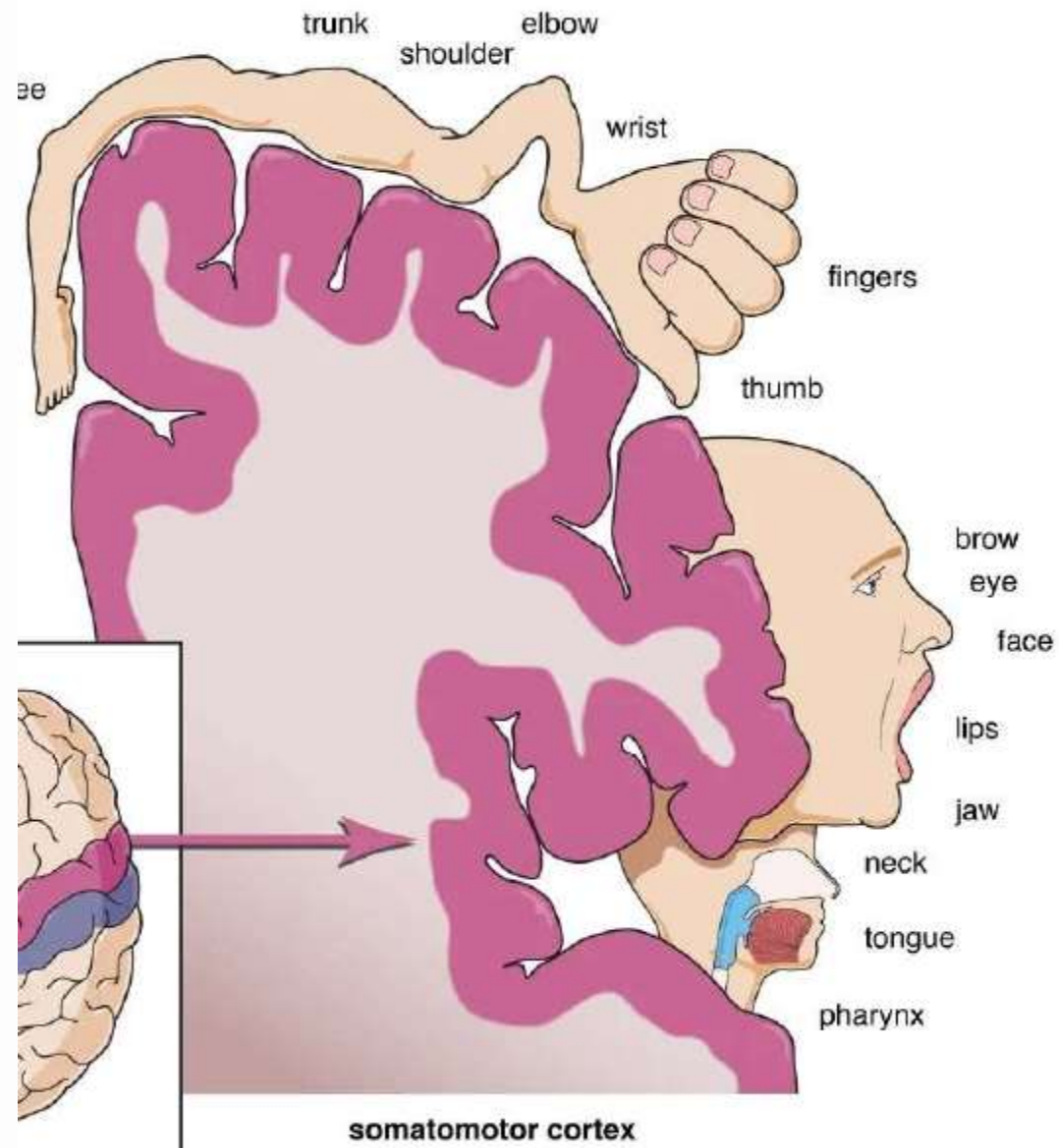
- Depending upon the topographical distribution
 - Monoplegia
 - Diplegia
 - Hemiplegia
 - Quadriplegia
- Depending upon the muscle tone
 - Spastic CP
 - Dyskinetic CP
 - Ataxic CP
 - Mixed CP



Spastic CP

- Most common
- 70-80 % cases
- Cerebral cortex gets affected
- Voluntary control disturbed
- Scissoring gait
- Increase deep tendon reflex
- Clonus is present
- Hypertonia
- Weakness
- Pathological reflexes seen
- Have mainly 3 subtypes – Quadriplegic, Diplegic, Hemiplegic





Dyskinetic CP

- Second most common
- Due to basal ganglia injury
- Unwanted motor movements seen
- Clinical diagnosis occurs generally within 18 months

Ataxic CP

- Due to cerebellum injury
- Least common
- Generally diplegic
- Wide BOS is seen
- Hypotonia emerges



ASSESSMENT

History

- Pre-Natal
 - Mother's age
 - Drugs used during pregnancy
 - Any infection during pregnancy
 - Any medical condition of mother
 - Any history of abortions or ectopic pregnancy
- Peri-Natal
 - Normal or Pre-term delivery
 - Labor duration
 - H/O Asphyxia
 - Birth cry



History

- Post-Natal
 - APGAR Score
 - Trauma
 - Any diseased condition
 - Nutritional habits
 - Any surgical condition
 - Any medical history



Observation

- Child alertness
- Communication
- Postural alignment
- Deformity
- Involuntary movements
- Scar
- Tremor



Examination

- Orientation
- Vision
- Hearing
- Speech
- Cranial Nerve Assessment
- Sensory examination
- Reflex assessment
 - Flexor withdrawal reflex
 - Palmar and plantar grip
 - Sucking and rooting reflex
 - Tonic labyrinthine reflex
 - ATNR
 - STNR



Examination

- Tone
- Muscle weakness
- Voluntary control
- Coordination
- Balance
- ROM
- GMFCS



MANAGEMENT

PT Management

- CP patients demonstrate a variety of issues and defects throughout their life
- Four stage of treatment is given to the CP Patients from infancy to childhood
- The four stages are :
 1. Early intervention
 2. Pre-School
 3. School age and adolescence
 4. Adulthood



Early intervention

- Positioning the child in supine promote head orientation in midline
- Reaching in supine position
- Positioning of infant in prone
- Active head lifting in prone
- Active extension in prone
- Infant should not sleep in prone
- Sitting and standing
- Flexed posture facilitates feeding
- Deeper respiration can also be encouraged by applying slight pressure at the thorax/abdomen prior to inspiration
- Gentle ROM
- Functional activities
- Reaching



Early intervention

- Take proper precaution because dislocation can occur
- Trunk flexion on physio ball
- Trunk extension on physio ball
- Sit to stand
- Stretching
- Constraint Induced Movement Therapy
- Splinting
- Task oriented approach



Pre School

- Major emphasis during this period is to promote mobility and functional independence
- Activities include promoting postural reactions to improve head and trunk control
- Teaching transitions and practice of self care skills
- If the child didn't achieved upright orientation and mobility during the early intervention, now it's time to make a concentrated effort to make them do so.
- Assistive device or Orthotic device can be used
- We should also start focusing on the ambulation of the child
- Gait difficulties include limited postural reactions during weight shifting, decreased single limb and double limb support time
- If child can not support body weight in standing or during sit to stand transition, then a part of the child body weight is given on extended arms and then child practices coming to stand.

Pre School

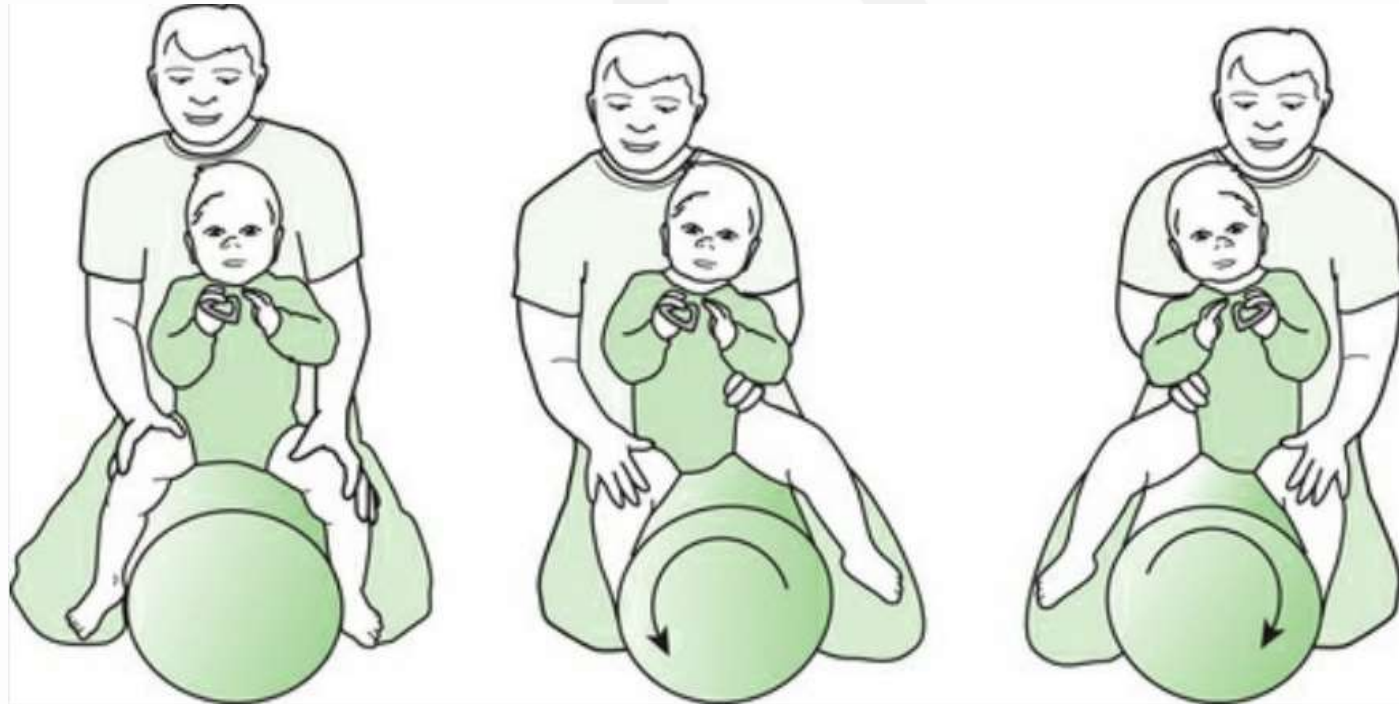


Pre School



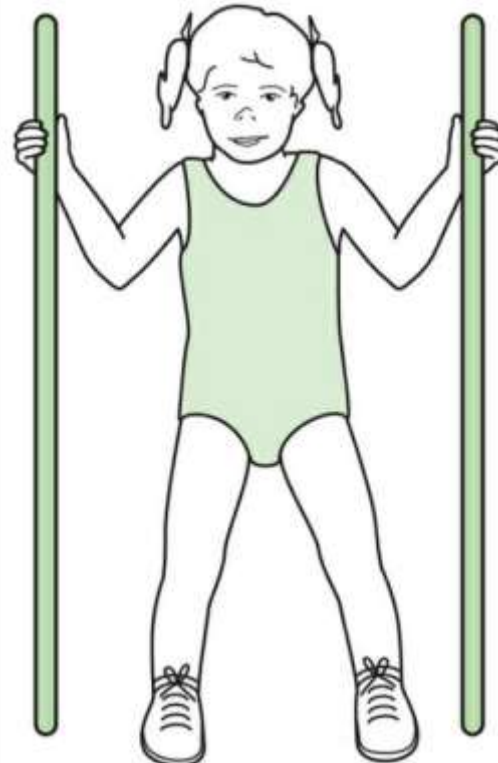
Pre School

- Trunk lateral bending may result in lower extremity separation as during the weight shifting, opposite leg is automatically abducted



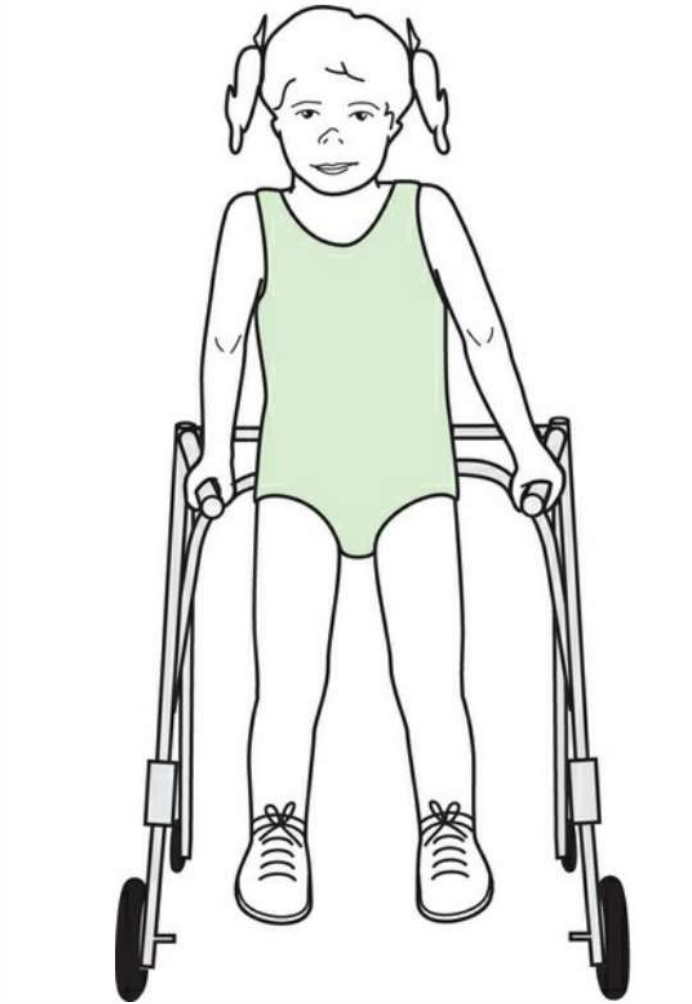
Pre School

- Pushing a toy and weight shifting
- As the child maintains the time for double limb support, progression can be made by stepping over an object or stepping up and down on stairs
- Having a child hold on to vertical poles decreases the amount of support and facilitates upper trunk extension



Pre School

- Many children can benefit from using a type of assistive device Rolling reverse walker during gait training



School age and adolescence/Adulthood

- Here main focus is to safeguard all the gains in the previous two stages
- Apart from the therapist, family and environment makes a great influence here.
- Remember 5 F's
 - Function
 - Family
 - Fun
 - Fitness
 - Friends
- Psychological management can also be given at this stage if needed
- We should focus on
 - Continue independent mobility and ADL
 - Community and social interaction

Spastic CP Management

Management

- Treatment focuses on mobility in all possible posture and transition between these postures
- Stretching of tight structures
- Area prone to more tightness in quadriplegic case
 - Shoulder adductors
 - Elbow flexor
 - Wrist flexor
 - Finger flexor
- Area prone to more tightness in diplegic case
 - Hip flexor
 - Hip adductor
 - Knee flexor
 - Ankle plantar flexor

Management

- ROM Exercises
- Weight bearing exercise
- Weight shifting
- Slow and rhythmic rocking
- Active trunk rotation
- Reaching in supine
- Trunk flexion and extension on physio ball
- Rolling on physio ball
- Gait Training
- Positioning
- Orthoses
 - AFO
 - Supramalleolar orthosis



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Positioning

- **Supine**
 - **Advantage**
 - Can begin early wt bearing through lower extremity when knees flexed and feet are flat on the surface
 - **Disadvantage**
 - Can be disorienting because it is associated with sleeping
 - Level of arousal will decrease
- **Side lying**
 - **Advantage**
 - Best positing to promote functional activity like rolling and transition
 - **Disadvantage**
 - Shortening of upper trunk muscles may occur if the child is always positioned on same side

Positioning

- **Prone**
 - **Advantage**
 - Can begin early wt bearing through upper extremity in prone on elbow or prone on arm
 - Stretch hip and knee flexors
 - **Disadvantage**
 - Breathing can be difficult
- **Sitting**
 - **Advantage**
 - Promotes active head trunk control
 - Weight bearing
 - Side sitting promotes trunk rotation
 - Helpful for performing functional activity with U/E

Positioning

- Sitting
 - Disadvantage
 - Child need support on trunk
 - Child unable to maintain neutral position / extension of trunk
- Quadruped
 - Advantage
 - Weight bearing
 - Trunk works anti gravity
 - Disadvantage
 - Posture is difficult to maintain because of the influence of STNR

Positioning

- Kneeling
 - Advantage
 - Provide stretch to hip flexors
 - Hip and pelvic control can be developed
 - Can further help in transition between different postures
 - Disadvantage
 - Posture is difficult to maintain

